ABC Parameter Reference — TNBC Lung Metastasis ABM

Integrated from all three papers

Paper codes used throughout:

- **[H22]** Hongu et al. 2022, *Nature Cancer* — endothelial & macrophage niche
- **[P20]** Pein et al. 2020, *Nature Communications* — fibroblast niche & IL-1 axis
- **[IR18]** Insua-Rodríguez et al. 2018, *EMBO Mol Med* — JNK signalling & ECM

TABLE 1 — Parameters to infer with ABC

These 10 parameters are free in every simulation run. Prior bounds are justified directly from paper data.

Parameter	Symbol	Prior	Lower	Upper	Paper figure	Observed constraint
JNK+ division prob	`divProbP`	Uniform	0.02	0.18	[IR18] Fig.1H	MDA-LM2 doubling ~24h in vivo; ~0.1/step at 30 min/step
JNK+ death prob	`dieProbP`	Uniform	0.005	0.06	[IR18] Fig.1H	Death << division for growing micromet; JNKi → no growth change in culture
JNK- division prob	`divProbN`	Uniform	0.01	0.12	[IR18] Fig.1	JNK- cells grow slower; must satisfy divProbN < divProbP
JNK- death prob	`dieProbN`	Uniform	0.005	0.05	[IR18] Fig.1	Similar death rate to JNK+; no direct measurement
JNK+ → JNK- switch rate	`activProbPN`	Uniform	0.015	0.08	[IR18] Fig.1E/F	JNK+ fraction: ~50% day7 → ~15% day21; must drive this decline
JNK- → JNK+ switch rate	`activProbNP`	Uniform	0.002	0.025	[IR18] Fig.1E/F	Asymmetric; must be << activProbPN so fraction does not rebound
Fibroblast activation prob	`activProbF`	Uniform	0.001	0.05	[P20] Fig.1f	100-fold fibroblast expansion over 3 weeks; [P20] Fig.6d: IL-1 KD → 50% reduction
Activated CAF division prob	`divProbFP`	Uniform	0.01	0.15	[P20] Fig.1f	Must reach ~10^6 from ~10^4 in 21 days
Macrophage activation prob	`activProbM`	Uniform	0.005	0.08	[H22] Fig.6e/f	~77% of nodule macrophages perivascular; interstitial macs 2–3x increase

Parameter	Symbol	Prior	Lower	Upper	Paper figure	Observed constraint
EC activation prob	`activProbE`	Uniform	0.005	0.08	[H22] Fig.1b	EC+ nodule %: 20% wk1 → 45% wk2 → 80% wk3

TABLE 2 — Parameters fixed from literature

These do not vary across ABC runs. Values are set at biologically defensible midpoints from the literature.

Parameter	Symbol	Fixed value	Source	Justification
JNK+ migration prob	`migrProbP`	0.10	[IR18] Fig.3M	JNKi reduces Matrigel invasion ~10x; migrProbP ≈ 10 × migrProbN
JNK- migration prob	`migrProbN`	0.01	[IR18] Fig.3M	Lower bound from invasion assay
Inactive fibroblast division	`divProbFN`	0.010	[LIT]	Lung fibroblast basal turnover ~7 days
Inactive fibroblast death	`dieProbFN`	0.008	[LIT]	Consistent with ~7 day half-life
Activated CAF death	`dieProbFP`	0.012	[LIT]	Slightly higher turnover when activated
Fibroblast migration	`migrProbF`	0.000	Model	Fibroblasts do not migrate in this model
Inactive macrophage division	`divProbMN`	0.005	[LIT]	Tissue macrophage slow proliferation
Inactive macrophage death	`dieProbMN`	0.008	[LIT]	Interstitial mac half-life ~1–2 weeks
Activated macrophage division	`divProbMP`	0.030	[H22] Fig.6f	Interstitial macs expand 2–3x; drives this
Activated macrophage death	`dieProbMP`	0.015	[LIT]	Higher turnover when activated
Mac inactivation rate	`activProbMP`	0.020	Model	Balances activated fraction at ~77%
Macrophage migration	`migrProbM`	0.100	[LIT]	Macrophages faster than tumour cells
Inactive EC division	`divProbEN`	0.003	[LIT]	Quiescent EC slow turnover (weeks)
Inactive EC death	`dieProbEN`	0.005	[LIT]	Quiescent EC basal death

Parameter	Symbol	Fixed value	Source	Justification
Activated EC division	`divProbEP`	0.040	[H22] Fig.1d	EC count doubles with tumour burden [H22 Fig.1d R~0.9]
Activated EC death	`dieProbEP`	0.008	[LIT]	Slightly higher than inactive
EC migration	`migrProbE`	0.000	Model	ECs form fixed tubes in this model
Lung cell division	`divProbL`	0.002	[LIT]	Alveolar epithelium slow turnover
Lung cell death	`dieProbL`	0.002	[LIT]	Steady-state homeostasis

TABLE 3 — Summary statistics for ABC calibration

All statistics are computed from `DetailedCounts{runId}.txt` *snapshots. Observed values are extracted directly from paper figures.*

Statistic	Symbol	ABM computation	Observed value	SD (from paper)	Paper figure	Weight ($1/\sigma^2$)
Tumour log-growth wk1→wk2	S[0][0]	log(tumJNKp+tumJNKn at step 960 / at step 480)	log(5.0) ≈ 1.61	0.40	[H22] Fig.1d	6.25
Tumour log-growth wk1→wk3	S[0][1]	log(tumJNKp+tumJNKn at step 1440 / at step 480)	log(20.0) ≈ 3.00	0.40	[H22] Fig.1d	6.25
Fibroblast log fold-expansion	S[1]	log((fibroAct+fibroInact at step 1440) / initial fibro count)	4.60 (≈100-fold)	0.50	[P20] Fig.1f	4.00
EC activated fraction wk1	S[2][0]	ecActive / (ecActive+ecInact) at step 480	0.20	0.15	[H22] Fig.1b	44.4
EC activated fraction wk2	S[2][1]	ecActive / (ecActive+ecInact) at step 960	0.45	0.15	[H22] Fig.1b	44.4
EC activated fraction wk3	S[2][2]	ecActive / (ecActive+ecInact) at step 1440	0.80	0.15	[H22] Fig.1b	44.4
Macrophage activation fraction	S[3]	macActive / (macActive+macInact) at step 1440	0.77	0.10	[H22] Fig.6e	100.0
Fibroblast–tumour correlation	S[4]	Pearson r between fibroblast count and tumour count across steps 480–1440	0.97	0.03	[P20] Fig.1d/e	1111.0
JNK+ fraction at day 7	S[5][0]	tumJNKp / (tumJNKp+tumJNKn) at step 480	0.50	0.12	[IR18] Fig.1F	69.4
JNK+ fraction at day 21	S[5][1]	tumJNKp / (tumJNKp+tumJNKn) at step 1440	0.15	0.12	[IR18] Fig.1F	69.4

Statistic	Symbol	ABM computation	Observed value	SD (from paper)	Paper figure	Weight ($1/\sigma^2$)
Nodule size fold-change wk1→wk3	S[6]	log(mean nodule radius at step 1440 / at step 480)	log(10) ≈ 2.30	0.35	[P20] Fig.1b; [H22] Fig.1c	8.16

Note on weights: weights = $1/\sigma^2$ where σ is the estimated SD from paper figures. S[4] (fibroblast–tumour correlation) has a very tight SD because R~0.97 is reported across many data points; it acts as a strong regulariser. If this over-constrains the posterior, relax to σ=0.10.

TABLE 4 — Validation statistics (not used in calibration)

Run these AFTER calibration to check biological plausibility of the posterior. Each is a perturbation experiment matching a specific paper knockout/inhibition figure.

Validation test	How to run	Expected ratio	Paper figure	What it tests
Fibroblast knockout	Set `activProbF=0`, `divProbFP=divProbFN`; run 100 posterior sims	tumour count wk3 ≈ 50% of control	[P20] Fig.6d: shIL1A/B → ~50% reduction	Whether `activProbF` and `divProbFP` jointly capture fibroblast contribution
JNK inhibition	Set `divProbP=divProbN`, `migrProbP=migrProbN`, `activProbPN→0`; run 100	tumour count wk3 ≈ 35% of control	[IR18] Fig.1L: JNKi → ~35% photon flux	Whether JNK+ advantage is correctly encoded
Macrophage depletion	Set `activProbM=0`, `divProbMP=divProbMN`; run 100	tumour count wk3 ≈ 50% of control	[H22] Fig.6l: clodronate → ~50% reduction	Whether macrophage-EC-tumour axis works
EC activation block	Set `activProbE=0`; run 100	EC activated fraction stays <5% at wk3	[H22] Fig.1b (implicit)	Whether EC activation requires macrophage signal
Chemo step	Run steps 2899-2900 of `StepCellRChemo()`; check JNK+ fraction	JNK+ fraction rises from ~15% to >80%	[IR18] Fig.5D/E: PAX → >80% p-c-Jun+	Whether chemo correctly drives JNK re-activation

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TABLE 5 — Identifiability assessment

Which parameters can actually be separated given your available data?

Parameter	Constraining statistic	Identifiable?	Risk	Notes
`divProbP`	S[0]: tumour log-growth trajectory	Yes — primary	Medium	Confounded with `dieProbP`; jointly constrained
`dieProbP`	S[0] + S[5]: JNK+ fraction decline	Yes — primary	Medium	Decline needs death to balance; use both
`divProbN`	S[0]: growth slower for JNK-	Partial	High	Hard to separate from `dieProbN` without JNK-specific data
`dieProbN`	S[0] + S[5]	Partial	High	Same issue; consider fixing ratio divProbN/dieProbN
`activProbPN`	S[5]: JNK+ fraction 50%→15%	Yes — primary	Low	This is the single best-constrained parameter
`activProbNP`	S[5]: fraction must not rebound	Yes	Medium	Constrained by absence of rebound; less informative than activProbPN
`activProbF`	S[1]: fibroblast fold-expansion + S[4]: fibro-tumour correlation	Yes — primary	Low	Two independent constraints from [P20]
`divProbFP`	S[1]: must reach 100-fold in 21 days	Yes — primary	Low	Directly sets expansion rate
`activProbM`	S[3]: macrophage activation fraction ~77%	Yes — primary	Low	Single tight observable
`activProbE`	S[2]: EC activation trajectory 20→45→80%	Yes — primary	Low	Three timepoints give strong constraint

Practical recommendation on `divProbN`/`dieProbN`: Given the limited data, consider fixing the JNK- net growth rate as `divProbN - dieProbN = 0.5 × (divProbP - dieProbP)` (JNK- grows at half the net rate of JNK+), and only freeing `divProbN` while computing `dieProbN = divProbN - 0.5*(divProbP-dieProbP)`. This reduces dimensionality from 10 to 9 free parameters without losing much biological information.

TABLE 6 — Observed quantitative values extracted from all three papers

Observable	Value	Uncertainty	Paper	Figure
EC+ nodule fraction at week 1	20%	±15%	[H22]	Fig.1b
EC+ nodule fraction at week 2	45%	±15%	[H22]	Fig.1b
EC+ nodule fraction at week 3	80%	±15%	[H22]	Fig.1b
GFP+ cancer cell count fold-change wk1→wk3	~20×	±30%	[H22]	Fig.1d
EC count fold-change wk1→wk3	~2×	±25%	[H22]	Fig.1d
EC–tumour count Pearson R	~0.90	—	[H22]	Fig.1d (implicit)
Perivascular macrophage fraction in nodules	76.8%	±10%	[H22]	Fig.6e
Interstitial macrophage fold-increase (MDA-LM2)	~2–3×	±40%	[H22]	Fig.6f
Macrophage depletion → tumour colonization reduction	~50%	±20%	[H22]	Fig.6l
Fibroblast count (healthy lung)	~10^4–10^5	—	[P20]	Fig.1f
Fibroblast count (macrometastasis)	~10^6	—	[P20]	Fig.1f
Fibroblast fold-expansion micro→macro	~100×	±50%	[P20]	Fig.1f
Fibroblast–tumour burden Pearson R (MDA)	0.974	—	[P20]	Fig.1d
Fibroblast–tumour burden Pearson R (MDA-LM2)	0.954	—	[P20]	Fig.1e
IL-1 KD → lung colonization reduction	~50%	±20%	[P20]	Fig.6d

Observable	Value	Uncertainty	Paper	Figure
CXCR3i → lung colonization reduction	~50–60%	±20%	[P20]	Fig.9d/e
Nodule size at week 1 (MDA-LM2)	~30–50 μm	±50%	[P20]	Fig.1b
Nodule size at week 3 (MDA-LM2)	~200–400 μm	±50%	[P20]	Fig.1b
p-c-Jun+ cells in micrometastases (day 7)	~50%	±12%	[IR18]	Fig.1F
p-c-Jun+ cells in macrometastases (day 21)	~15%	±12%	[IR18]	Fig.1F
p-c-Jun+ cells in lung mets vs primary tumour	~40–60% vs ~5–10%	—	[IR18]	Fig.1C/D
JNKi → lung colonization reduction	~65%	±20%	[IR18]	Fig.1L
JNKi → tumour initiation reduction at 500 cells	~60%	—	[IR18]	Fig.3L
JNKi → Matrigel invasion reduction	~10×	—	[IR18]	Fig.3M
Chemo → p-c-Jun+ fraction	<20% → >80%	—	[IR18]	Fig.5D/E
SPP1+TNC high → lung met-free survival HR	6.14	P=0.001	[IR18]	Fig.4L
CXCR3+ cell fraction (monolayer)	3–12%	—	[P20]	Fig.8a
CXCR3+ cell fraction (spheres)	~25–40%	—	[P20]	Fig.8a

TABLE 7 — Chemo parameter modifications (steps 2899–2900)

These multiplicative modifications apply inside `tumorCellChemo()`. All are fixed — not inferred — because they are directly specified by [IR18].

Modification	Factor	Applied to	Paper figure	Mechanism
JNK+ death increase	2.0 × `dieProbP`	JNK+ cells under chemo	[IR18] Fig.5D/E	Chemo cytotoxicity; JNK+ partially protected via SPP1/TNC
JNK+ → JNK- switch suppression	0.05 × `activProbPN`	JNK+ cells under chemo	[IR18] Fig.5D/E	Chemo locks cells in JNK+ state (c-Jun phosphorylation)
JNK- death increase	5.0 × `dieProbN`	JNK- cells under chemo	[IR18] Fig.5	JNK- cells more chemo-sensitive (no SPP1/TNC protection)
JNK- → JNK+ switch increase	5.0 × `activProbNP`	JNK- cells under chemo	[IR18] Fig.5D/E	Stress drives JNK activation (chemo as stress signal)

TABLE 8 — Time-step calibration

Connecting ABM steps to biological time, needed to read snapshot steps correctly.

ABM steps	Approximate biological time	Basis
1 step	~30–60 min	TNBC migration ~20 μm/h; grid spacing ~20 μm; 1 step = 1 move
480 steps	~10–20 days (≈ week 1-2)	Used as "week 1" snapshot for S[2][0] and S[5][0]
960 steps	~20–40 days (≈ week 2-3)	Used as "week 2" snapshot for S[2][1]
1440 steps	~30–60 days (≈ week 3)	Primary calibration target for all statistics
2100 steps	~43–87 days (≈ day 21 biological)	Used for JNK+ fraction S[5][1] [IR18 Fig.1F day 21]
2899 steps	End of pre-chemo phase	Chemo introduced at step 2899

Important caveat: The step-to-time mapping is uncertain by a factor of ~2. The key observables are all *ratios* (JNK+ fraction, EC activation fraction, fibroblast fold-expansion) or *log-ratios* (tumour growth), which are robust to this uncertainty. The absolute step numbers above should be treated as approximate. A dedicated sensitivity analysis varying the step-to-time mapping by ±50% is recommended before publishing.